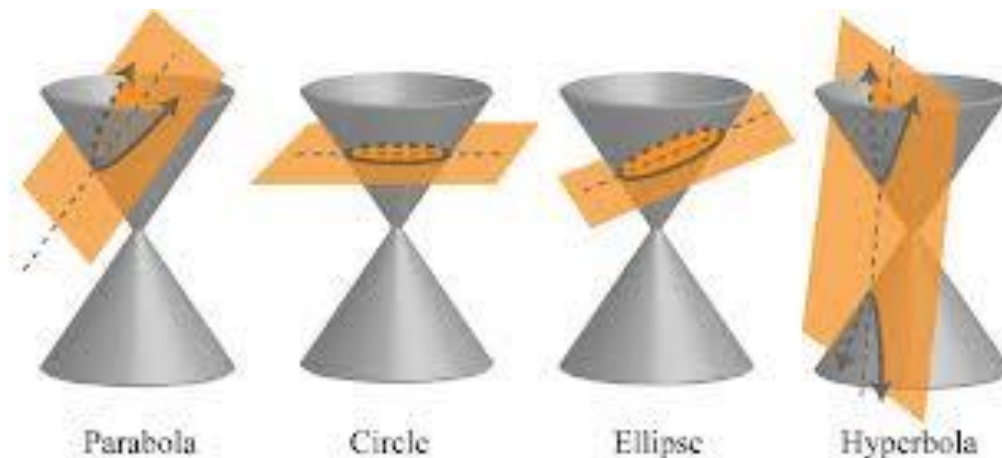


5.4.2 Conic Sections (Ellipses)

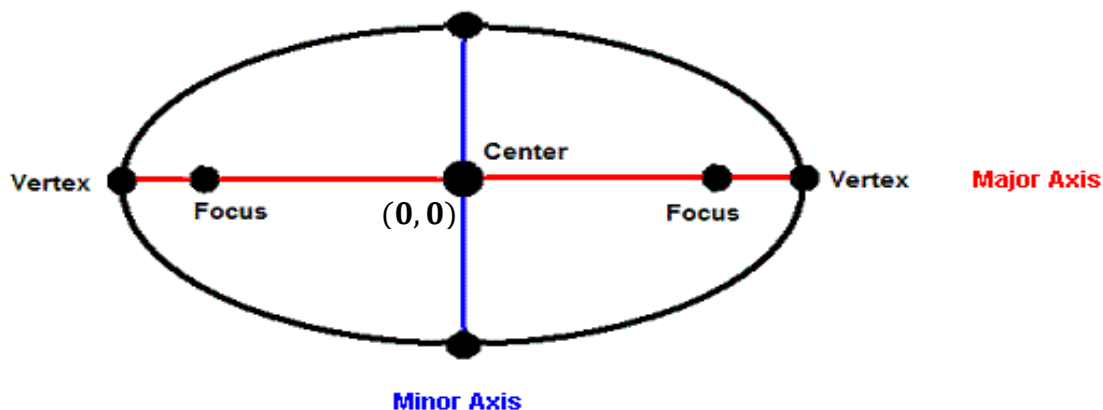
If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



Definition: An **ellipse** is the set of all points in a plane such that the sum of their distances from two fixed points is constant. Each fixed point is called a **focus** (plural **foci**) of the ellipse.

The Equation of an Ellipse: (with a *horizontal* major axis and center at the origin)

The ellipse with center at the origin and equation



The Equation of an Ellipse: (with a *vertical* major axis and center at the origin)

The ellipse with center at the origin and equation

Example 1: Sketch a graph and identify the foci of the ellipse:

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

Example 2: Sketch a graph and identify the foci of the ellipse:

$$11x^2 + 3y^2 = 66$$

Example 3: Write the equation of an ellipse with foci $(0, \pm 4)$ and vertices $(0, \pm 6)$.

Fact: The equation representing a conic that has been shifted h units horizontally and k units vertically can be formed by replacing any x 's in the original equation with $x - h$ and replacing any y 's in the original equation with $y - k$.

Example 4: Sketch a graph and identify the foci of the ellipse: $9x^2 - 54x - 19 = 100 - 50y - 25y^2$

The Circle

Definition: A circle is a set of all points in a plane such that their distance from a point is constant.

Equation for a Circle

The standard equation of a circle with center (h, k) and radius r ($r > 0$) is

Note: By dividing both sides of this equation by r^2 we obtain $\frac{(x-h)^2}{r^2} + \frac{(y-k)^2}{r^2} = 1$. So clearly a circle is an ellipse with major and minor axes of the same length.

Example 5: Write the equation of a circle that has center $(4, 5)$ and passes through the point $(-1, 2)$.

Example 6: Determine the center and radius and sketch a graph of the circle given by the equation:

$$x^2 + 4x + y^2 - 2y = 11$$